



SJB INSTITUTE OF TECHNOLOGY

(An Autonomous Institute under
Visvesvaraya Technological University, Belagavi)

BENGALURU-60



TITLE PAGE

SMART INDIA HACKATHON 2026

- Problem Statement ID –
- Problem Statement Title-
- Title-
- Name-





SJB INSTITUTE OF TECHNOLOGY

(An Autonomous Institute under
Visvesvaraya Technological University, Belagavi)

BENGALURU-60



DRISHTI

PROPOSED SOLUTION



MoSJE Oversight & Web Dashboard
Define Policy, Monitor



Central AI Hub & Automation
Data Process, Random Assign



Field App & Live Feeds
Capture Geo- Evidence, Live VC



Analytics & Insights Detect
Anomalies, Verify Attendance

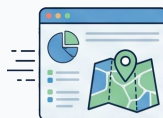
HOW IT ADDRESSES THE PROBLEM



Eliminates Fake Reporting
Tamper-proof Data



Prevents Collusion
Random Assignments

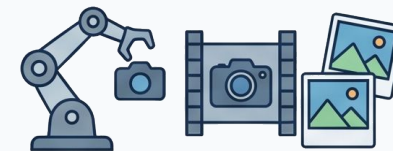


Overcomes Bottlenecks
Efficient Scalability



Automates Compliance
24/7 AI Auditing

INNOVATION & UNIQUENESS



Lightweight AI
Efficient, Low-Bandwidth Vision



Zero-Trust Evidence
Hardware-Verified Data



Unified Ecosystem Centralized
Monitoring Platform



SJB INSTITUTE OF TECHNOLOGY

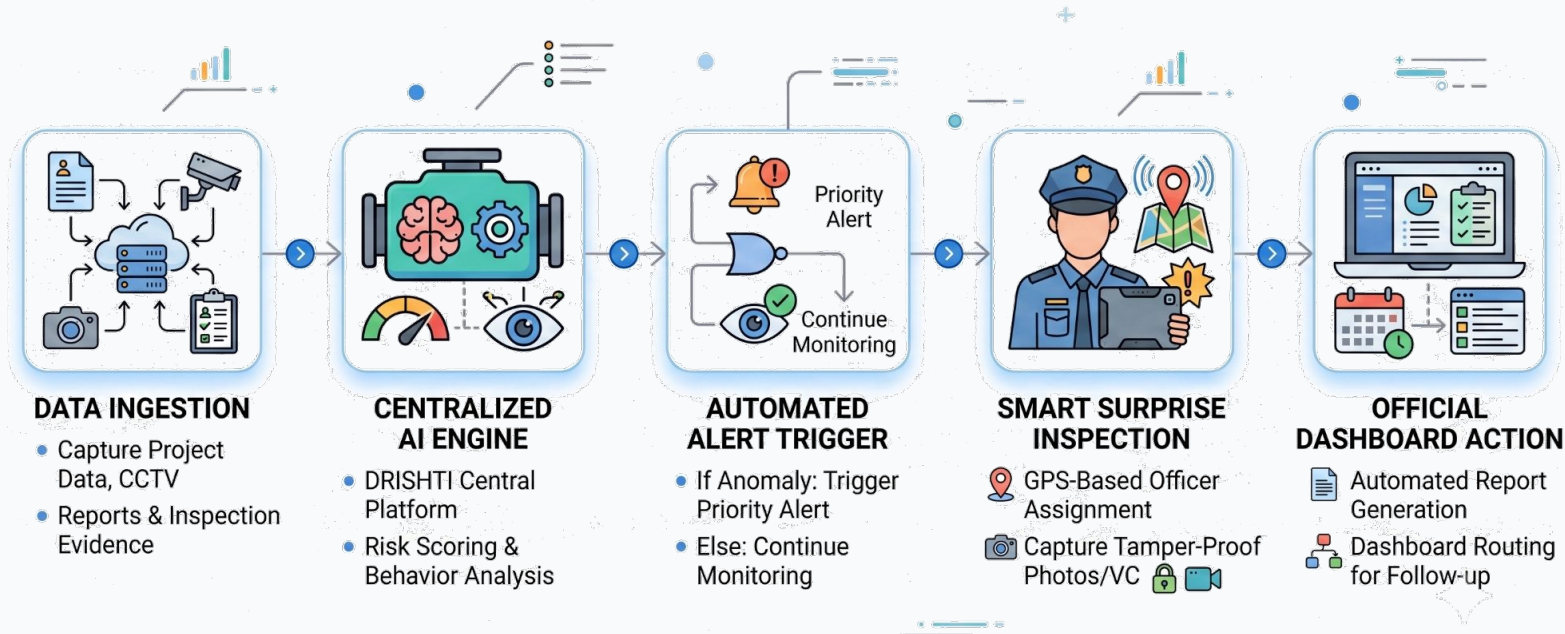
(An Autonomous Institute under
Visvesvaraya Technological University, Belagavi)

BENGALURU-60



TECHNICAL APPROACH

METHODOLOGY & FLOW



TECH STACK



Technical Approach — DRISHTI

Tech Stack & Tools

Frontend & Maps

Flutter / React Native (Cross-platform), GPS + Google Maps API

Backend & DB

Python (FastAPI) + Node.js APIs, PostgreSQL & Firebase

AI & Computer Vision

OpenCV, YOLO, Scikit-learn (Anomaly, attendance & activity analytics)

Surveillance & Comm

RTSP/IP Camera, WebSockets (Live CCTV), WebRTC (Secure VC)

Security & Deployment

JWT, RBAC, AES-256 encryption, Docker + Cloud/On-Premise

Methodology Flow

1. Data Ingestion & Live Feeds

Capture Project Data, CCTV, Reports & Inspection Evidence.

2. Centralized AI Engine

DRISHTI Central Platform performs risk scoring & behavior analysis.

3. Automated Alert Trigger

If anomaly is detected, trigger priority alert; else continue monitoring.

4. Smart surprise Inspection

Assign nearby field officers via GPS. Capture tamper-proof photos/VC.

5. Official Dashboard Action

Automated report generation and dashboard routing for follow-up.

Prototype & Principle

Prototype Workflow

App → API Gateway → AI Engine → DB → Dashboard

Demonstrates role-based login, GPS-based allocation, evidence capture, anomaly alerts, and automated reporting.

Core Operating Principle

MONITOR



DETECT



VERIFY



REPORT



ACT

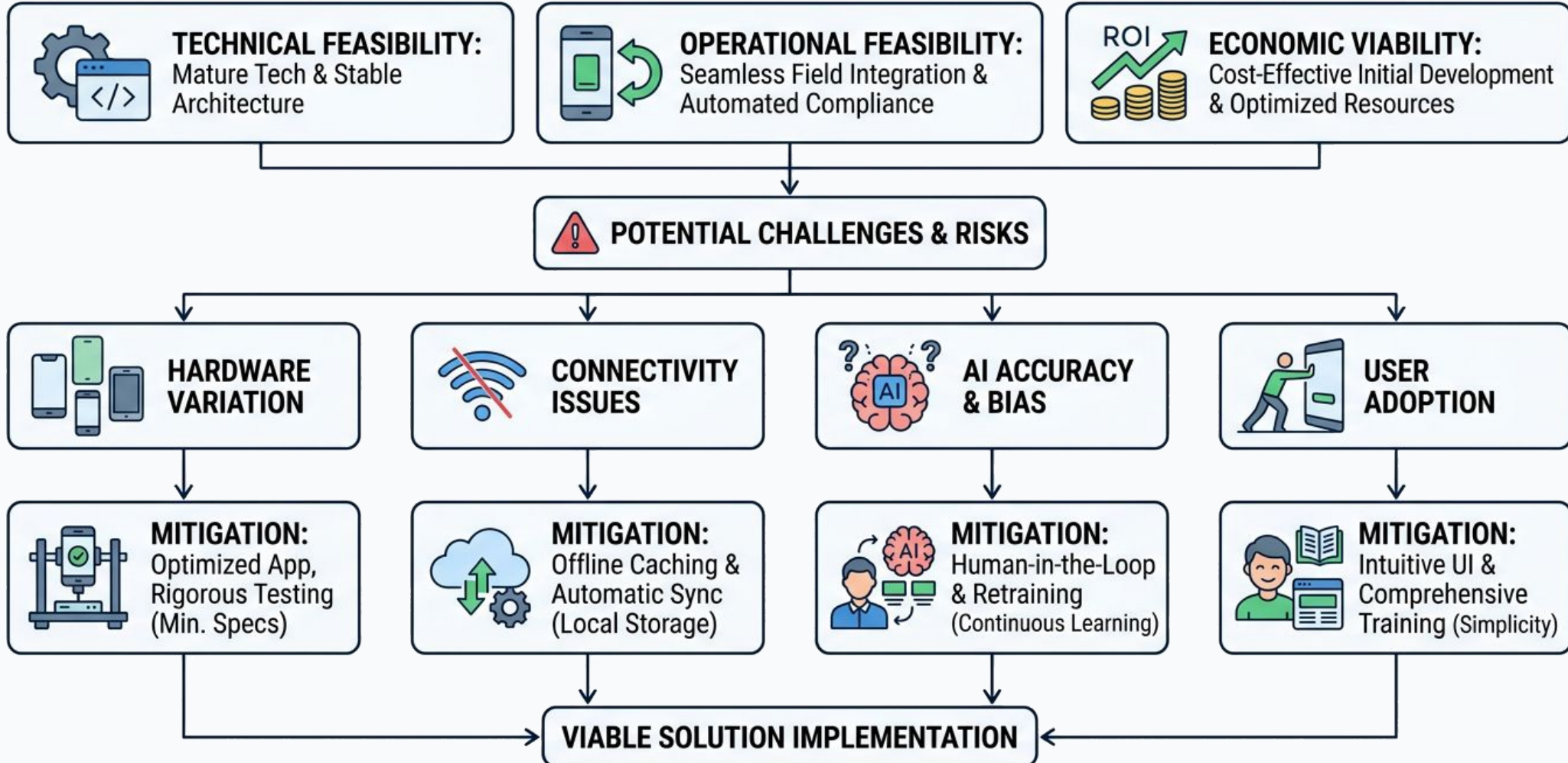


SJB INSTITUTE OF TECHNOLOGY

(An Autonomous Institute under
Visvesvaraya Technological University, Belagavi)
BENGALURU-60



FEASIBILITY & VIABILITY

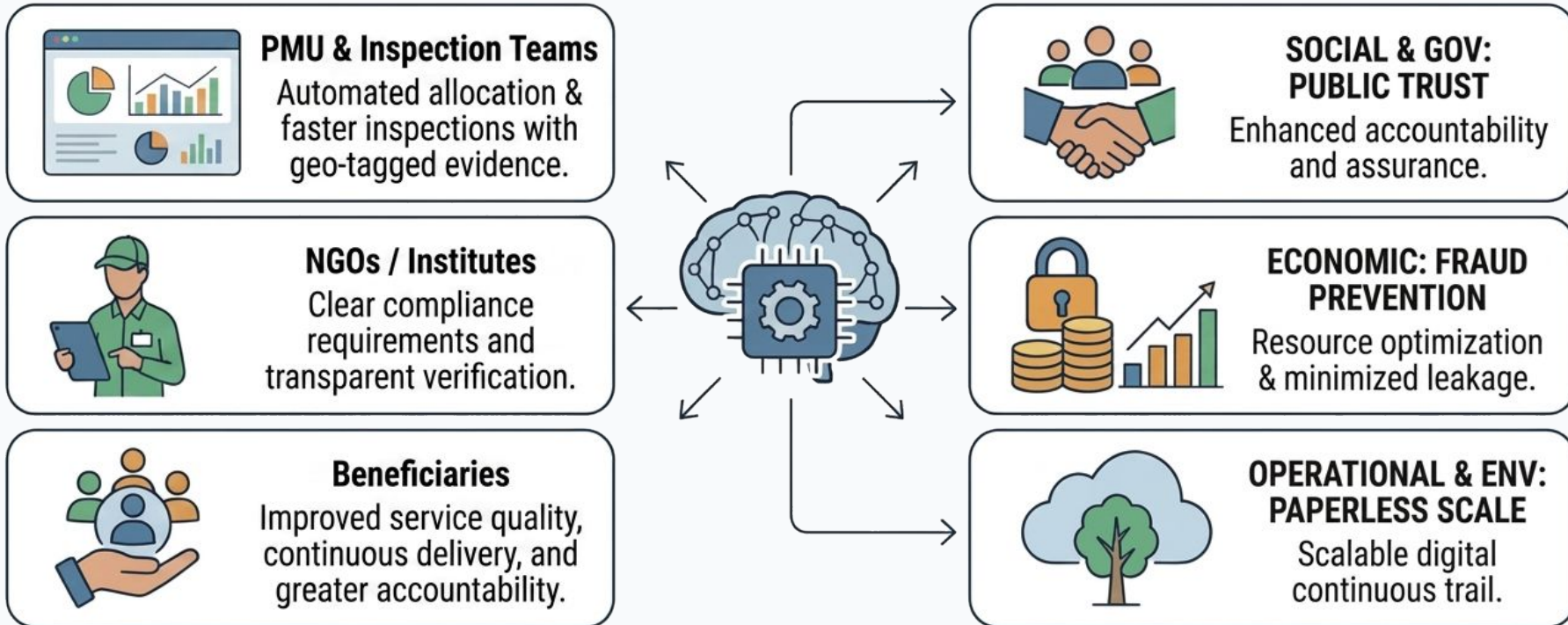




SJB INSTITUTE OF TECHNOLOGY

(An Autonomous Institute under
Visvesvaraya Technological University, Belagavi)
BENGALURU-60

IMPACT & BENEFITS



FROM:  REPORT-BASED INSPECTION → TO:  EVIDENCE-BASED CONTINUOUS VERIFICATION



SJB INSTITUTE OF TECHNOLOGY

(An Autonomous Institute under
Visvesvaraya Technological University, Belagavi)
BENGALURU-60



REFERENCES & RESEARCH — DRISHTI



Official Government References

Ministry of Social Justice & Empowerment

Official social welfare initiatives & policies

<https://socialjustice.gov.in/>

Dept of Social Justice & Empowerment

Program execution & welfare schemes monitoring

<https://socialjustice.gov.in/schemes>

Digital India Portal

Digital governance & service delivery standards

<https://digitalindia.gov.in/>

UIDAI Aadhaar & e-KYC

Secure beneficiary identity verification

<https://uidai.gov.in/>



Technical Documentation References

OpenCV Framework

Computer vision processing algorithms

<https://docs.opencv.org/>

Ultralytics YOLO Engine

Real-time object detection & site monitoring

<https://docs.ultralytics.com/>

WebRTC Protocol

Real-time low-latency video calls & checks

<https://webrtc.org/>

PostgreSQL DB Engine

Scalable relational database infrastructure

<https://postgresql.org/docs>